Four Ways to See Iris: Exploring Visualization Design Choices

A Cognitively-Driven Design Experiment

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Motivating Problem

Designing effective data visualizations involves making countless choices. Designers must make calculated decisions on color schemes, label placements, interactivity, and much more to communicate their underlying data effectively. Most of these choices are common, and designers often make them based on personal preferences and convention rather than empirical evidence.

Our project investigates how design choices help or hurt users' abilities to interpret data, and if their own personal preferences appear to vary or merge with the preferences of their peers. We focus on color scheme and label interactivity as design choices, since these are common features that users often notice first. This matters because small design choices can significantly affect whether a general audience will walk away from a visualization with the intended impression of the designers.

Data

For this project, we use the Iris dataset. The Iris dataset is a well-structured and simple dataset with sufficient variables for a diverse set of visualization types. It allows for clear classifications, grouping color schemes, and easy-to-understand measurements for labels. It is approachable for participants who may or may not have experience with visual design, and it is simple enough that experienced users will hopefully pay no attention to it in their analysis. Iris naturally supports many chart types. It has clear categorial data suited for pie and bar charts, and can be used for multivariate comparisons, which fit nicely into scatter and radar/spider charts. Demographically, Iris consists of four features: sepal length, petal length, sepal width, and petal width. It has 150 instances and is classified by three flower groups: Setosa, Versicolor, and Virginica.

Tasks and Insights

The visualizations will allow the participants to compare multiple versions of the same data and identify what kind of visualization best supports their understanding of the dataset. The participants will also evaluate how the visualization design choices affected their experience. They will be asked which visualization they preferred out of the set and why. This allows us to understand how color schemes and labels interactively influenced their comprehension of the data. This helps us draw insights into how

certain design choices improve accuracy and user confidence. It also allows us to see if users prioritize aesthetics or interactivity with the data visualizations.

Preliminary Visualization and Design Choice

Our idea is to show participants four versions of the same visualization. The underlying data will be the same in all four visualizations, but participants will have to choose between a color scheme and a level of interactivity for each visualization type. Below is our preliminary drawing for what one set of visualizations (pie charts) might look like.

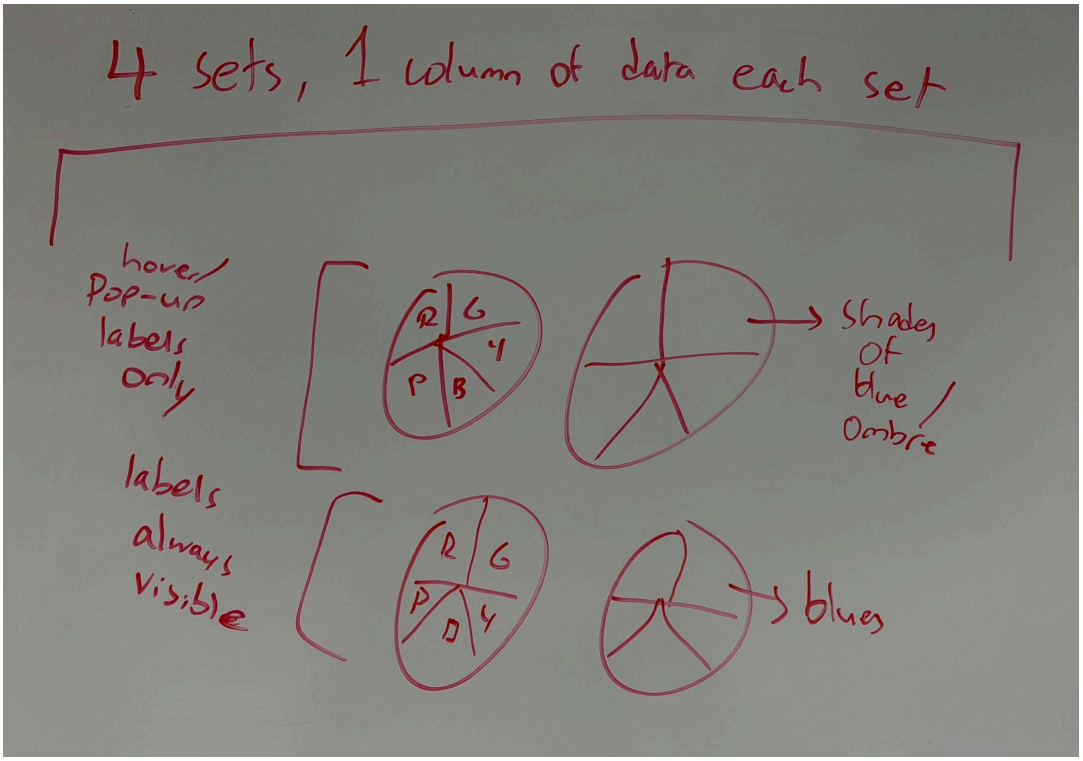

Fig. 1: Preliminary visualization illustrating one set of visualizations where the top two graphs feature different color choices and interactive labels, and the bottom two graphs feature different color choices and always visible labels.

Chosen Method: Cognitively-Driven Design

We chose the cognitively-driven design method as it is more experiment-focused and allows us to have greater control over the participants' experience during the study. There are two main variables that we control, and this design method allows us to explicitly track which changes in design elicit which responses from the participants. Our target question asks what visualization style (taking into account chart type, color scheme, and interactivity level) allows users to best understand the underlying data.

Independent Variables. Our experiment has two independent variables: color and interactivity. Colors will be controlled by splitting all charts into either full rainbow colors or shades of purple. We will also split graphs so half have interactive labels and the other half will display all information all the time. Thus, there are four versions of each visualization type, allowing for each combination of color scheme and interactivity level.

Dependent Variables. Our dependent variables are accuracy and similarity. Since all relevant information will be present in half of the charts, accuracy will be measured in users' ability to properly explain their reasoning for picking a favorite chart each round. In our follow-up discussion, we will compare users' responses to what charts they say they would've chosen themselves. Further details on this are in the "Experiment Details" section. Our second dependent variable, similarity, compares how similar the participants' responses are to each other. If there are any outliers, we will refer to that response's (anonymized) comments to learn more.

Control Variables. Our first control variable is the type/number of graphs used. We are presenting four types of visualizations (pie chart, bar chart, scatter plot, and spider plot)- four charts, each type (varying colors and interactivity). Our other control variables are the timeframe and group makeup. Everyone is coming in the same time block (the exam period) and has the same amount of time to evaluate the graphs.

Hypotheses. Before running our experiments, we developed two hypotheses. The first hypothesis is that rainbow colors will work better with pie and bar charts, while ombre colors will work better for scatter and spider plots. We make this prediction on the assumption that different categories warrant different colors (hence, rainbow colors emphasizing splits in pies and bars), while similar colors better show changes in data (ideal for scatter and spider plots). Our second hypothesis is that among all charts, users will prefer the interactive, hoverable labels over static charts that have all data present. When the number of categories increases, having all data always present might clutter the charts. Interactivity allows users to explore the data at their own pace, allowing them to make their own conclusions about the data.

Experiment Details

We will present participants with all four charts of a certain visualization type (bar, spider, etc.) at once. We will give them a few moments to explore the charts and then have them identify their preferred visualization among the four. Participants will also provide brief comments on why they chose their selected graph. We will use these comments to understand overall trends and the reasoning behind any outliers.

During the experiment, we passed around our laptops to each group, allowing them to explore the visualizations at their own pace. We had all participants explore pie charts first, then bar charts, scatter plots, and spider plots last. For each set, we gave participants a few minutes to explore, then asked them to select their favorite chart and leave their comments.

Once participants have viewed and voted on all sixteen visualizations, we will begin the second portion of our experiment. In this portion, participants will be encouraged to explore the whole Iris dataset at their own pace. After an adequate amount of exploration time, we will ask participants to describe what

visualization they would have chosen for each column. Since participants will have already explored four possible visualizations for each column in the dataset, they will have a preliminary understanding of what visualization features work and don't work for this dataset. This discussion will thus encourage deeper thinking about data visualizations and the methods used in designing them.

Open Science. The live deployment of the experiments is accessible at this GitHub Pages, and the survey at this Google Form.

Procedure

Participants will be introduced to the tasks outlined above and shown the four chart variants for each chart type. They will then complete a short Google Form asking which chart variant they prefer at each step, prompting them to describe what they observe in their selected charts.

After the form, we will give participants access to the Iris dataset and allow them to briefly explore it. We will then facilitate a short discussion by asking participants what design choices they themselves would have made, what features would have mattered to them, and why.

Analysis Plan

We noted which variants participants preferred and whether their preferences changed upon seeing the raw data. We had participants fill out a Google Form, which created a basic analysis of the responses. Our analysis was based on the default pie charts generated by Google Forms and the comments participants left, as well as a live discussion with the groups after the fact.

Analysis

The first set of visualizations we had participants view was pie charts. Below is a pie chart (yes, we see the irony of this, pie-ception) showing the response breakdown of participants' favorite pie charts.

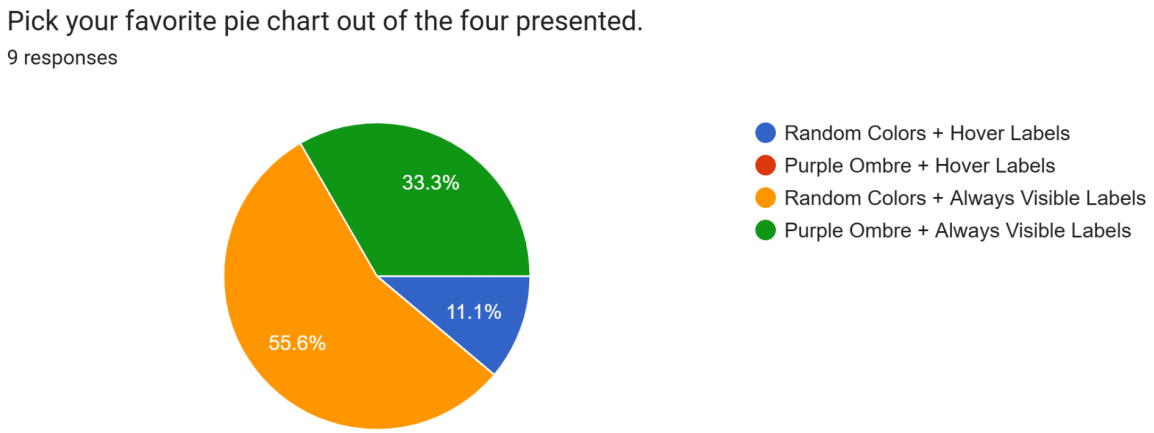

Fig 2: Participant responses showcasing their favorite pie charts.

Over half of the participants chose 'Random Colors + Always Visible Labels.' Almost 90% of the participants chose the 'Always Visible Labels.' The individual comments concluded that they liked seeing the labels. They're effective, easier to read, and you don't have to wait to hover over the pie. The participants were more split on the color. Some participants liked the differentiation of irises with random colors. Others thought it was intuitive for each slice to have the same hue since the graphs represent the Iris data.

The second set of visualizations we had participants explore was bar charts. Below is the pie chart that represents our participants' responses to their favorite bar chart.

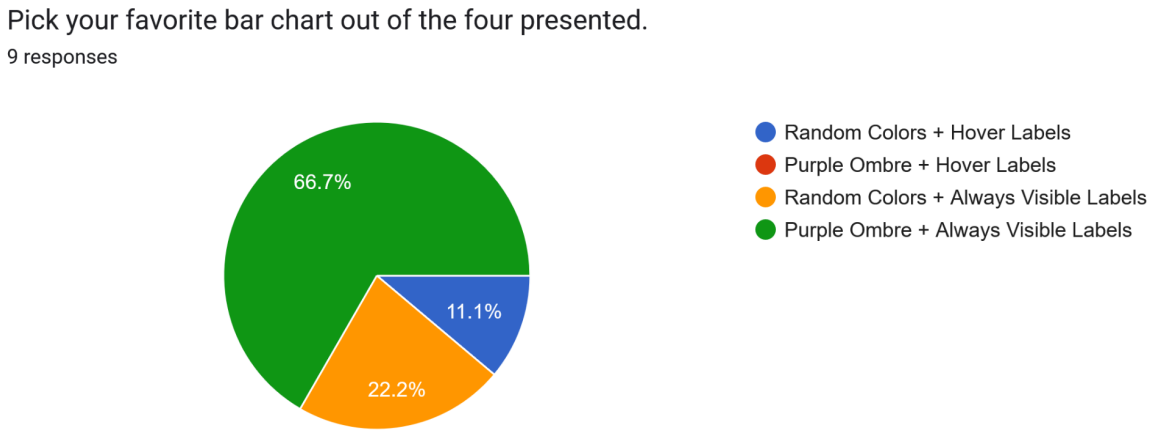

Fig 3: Participant responses showcasing their favorite bar charts.

Similar to the pie chart results, most users liked the 'Always Visible Labels' to explain the quantity of data for reasons outlined below Figure 1. However, the majority of users preferred the 'Purple Ombre'

colors to indicate ordered data; they also felt the darker colors showing greater quantities helped deepen their understanding of the graph.

The third set of visualizations we had participants explore is scatter plots. As the pie chart below shows, every single participant chose the same scatter plot as their favorite: 'Random Colors + Hover Labels'.

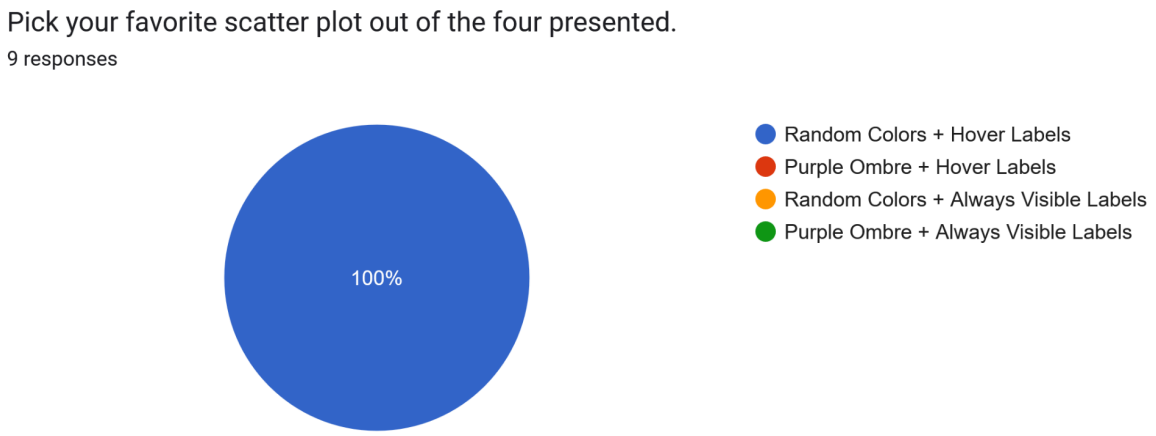

Fig. 4: Participant responses showcasing their favorite scatter plots.

Every participant chose 'Random Colors + Hover Labels.' Overwhelmingly, the participants noted that random colors made the distinction between iris flowers more notable. One participant said the purple shades made overlapping data harder to read. The participants also agreed that the 'Hover Labels' were much easier to read and made the data more readable. The 'Always Visible Labels' were cluttering the view of the graph and were not easy to interpret.

The final set of visualizations we had participants explore is spider plots.

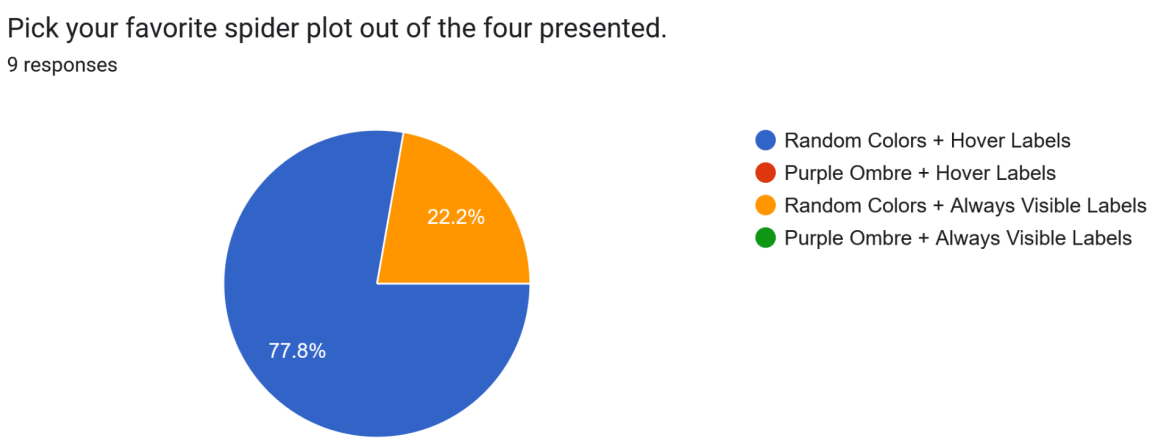

Fig. 5: Participant responses showcasing their favorite spider plots.

Participant responses for the spider plots were split among the two versions of the 'Random Colors' visuals. Comments discussed how the different colors make the labels easier to read (in both label versions) and make the shapes easier to differentiate. The majority of participants favored 'Hover Labels' over the "Always Visible Labels', stating that they're easier to understand and look cleaner.

Post-Survey Discussion. After the three groups finished the survey and explored the Iris dataset, we asked them two questions: (1) What feedback did they have on the visualizations they saw and their favorite vs. least favorite designs; (2) After viewing the designs and data, what would they have done differently to visualize the data. We recorded these results by hand and have transcribed them below.

For question one (1), all three groups (Groups 14-16) liked different design choices across a variety of visualization types. Group 15 mentioned their favorite way to visualize the clustering of the three iris groups was through the scatterplot; Groups 14 and 16 favored the spider charts for visualizing the average measurements of the three groups, and noted that the spider charts looked like flower petals. They went on to say that they like it when visualization design is tied to the underlying data, through common motifs like symbolism and color scheme.

For question two (2), Group 15 discussed how they would make visualizations for the Iris dataset that showed the variability of the length and width measurements for each category of irises. This would illustrate how petal length and width vary across the categories to get a sense of not only the center of the data, but the spread. Group 16 discussed how they have become "desensitized" to pie charts, and that they would have appreciated more low-fidelity visualizations with a greater variety of tested gestalts and more unique visualization types. They also asked if we took colorblindness and accessibility into consideration, as these could affect a user's perception of the data.

Discussion and Limitations

After analyzing all our results, we compare our data to our initial hypotheses. Our first hypothesis is that rainbow colors will work better with pie and bar charts, while ombre colors will work better with scatter and spider plots. This hypothesis was partially proven true, but was mostly false. Participants did prefer rainbow colors for pie charts, but picked the opposite color scheme compared to our expectations for the remaining visualization categories. Our participants preferred ombre colors for bar charts and random colors for both scatter and spider plots.

Our second hypothesis that users will prefer the interactive hoverable labels in all visualization categories was also only partially correct. Users preferred hoverable labels for scatter and spider plots, but preferred the always-present labels for pie and bar charts. From participant comments, we gather that the preference for hoverable labels is only when the visualizations have many other elements already present. Always visible labels in these visualizations distract from the overall bigger picture. However, in simpler visualizations (i.e., our pie and bar charts), the always visible labels were an essential component in understanding the data. There were fewer data points present, so the labels were added to the visualization rather than cluttering it.

We also aimed to measure how similar the responses were from the individual participants within a single graph. Overall, we found that for each graph, at least 89.9% of responses shared a single trait. For example, every participant preferred random colors in their spider charts, while they differed on label visibility. Additionally, we found that there was more variation in responses for the pie and bar charts, while the spider and scatter plots had unanimous consensus on using random colors.

Our study is small and uses a sample of classmates with similar visualization and data analysis skills. The experiment itself only tests two design dimensions and four chart types, making findings not able to be extended to other design choices or datasets. Subjective judgment from participants will introduce inconsistencies and may make results reflect personal bias rather than any ground truth. Finally, being the group nature of the project, the responses from the discussion portion risk social influence.

Future improvements of this experiment based on the post-survey discussion could include more ‘complex’ or less common graph visualizations. One participant noted that they were "desensensitized" to pie charts. Our experiment and hypotheses were specifically testing two gestalt principles. To expand the experiment, we could include more low-fidelity visualizations to test a greater variety of gestalts.

Conclusion

We developed an experiment that analyzed the effectiveness of different visualization qualities and gestalts across four visualization types to see if our preconceived notions of important features aligned with those of our peers in our community. We found that our hypotheses were not aligned with the results of our experiment, and that these hypotheses may have reflected our own biases or biases that existed within the small survey group, leading to interesting discussion and reflection. This project could be extended to more complex datasets, testing different design qualities or visualization types, and to different user personas. Overall, we found that designing engaging and effective visualizations requires multiple iterations of attempts, input from many personas, and still may not be effective based on our own lived experiences and personal preferences.